Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples		
Distributive Property Recall: $a(b+c)=$ $ab+ac$ or $a(b-c)=$ $ab-ac$	1. $7(x+4)$ $7x+28$	2. $2(b-3)$ $2b-6$	3. $-4(y+3)$ $-4y-12$
	4. $-5(m-2)$ $-5m+10$	5. $-(y-9)$ $-y+9$	6. $8(4-b)$ $32-8b$
	7. $-4(-w-10)$ $4w+40$	8. $5(2m-3)$ $10m-15$	9. $-3(2x-4)$ $-6x+12$
	10. $3(m+n)$ $3m+3n$	11. $x(y+4)$ $xy+4x$	12. $d(c-4)$ $dc-4d$
	13. $6a(b+2c)$ $6ab+12ac$	14. $2(x-y+8)$ $2x-2y+16$	15. $x(2y-5z+1)$ $2xy-5xz+x$
	16. $\frac{1}{4}(8x-4)$ $2x-1$	17. $\frac{1}{6}\left(\frac{3}{5}x+18\right)$ $\frac{1}{10}x+3$	18. $-\frac{1}{8}\left(\frac{4}{5}x-24\right)$ $-\frac{1}{10}x+3$
Distribute AND Combine!	To **simplify an expression** means to ensure there are **no parentheses** and **no like terms.** In order to do this, distribute first (if needed), then combine like terms.		
Examples	19. $8(2x-3)-6x$ $16x-24-6x$ $=\boxed{10x-24}$	20. $9(2k-4)-2(7k-12)$ $18k-36-14k+24$ $=\boxed{4k-12}$	
	21. $10-(y-6)-y$ $10-y+6-y$ $=\boxed{-2y+16}$	22. $6+8(4w-7)-(2w+1)$ $6+32w-56-2w-1$ $=\boxed{30w-51}$	
	23. $\frac{9}{4}\left(\frac{10}{3}x-4\right)-\frac{1}{4}(2x-8)$ $\frac{15}{2}x-9-\frac{1}{2}x+2$ $=\boxed{7x-7}$	24. $\frac{7}{8}\left(\frac{4}{5}x+16\right)+\frac{3}{5}\left(\frac{1}{2}x-10\right)$ $\frac{7}{10}x+14+\frac{3}{10}x-6$ $=\boxed{x+8}$	

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